

Chapter 3. Advanced Modular Arithmetic

Section 3.1. Orders

3.1.1 Properties of Orders

In this section, we introduce an important concept that will play a central role in our number theoretic explorations. Our starting point is the following question:

If a prime p divides $a^k - 1$, what can we say about a , k , and p ?

Recall *Fermat's Little Theorem*, which states that if $p \nmid a$, then

$$a^{p-1} \equiv 1 \pmod{p}$$

Thus, we already know that such an exponent k exists (for example, $k = p - 1$).

But is $p - 1$ always the smallest such exponent? The answer is no. For example,

$$2^3 \equiv 1 \pmod{7},$$

even though $p - 1 = 6$. In this case, the smallest such exponent is $k = 3$.

This naturally leads us to ask: what is the *smallest* positive integer k such that

$$a^k \equiv 1 \pmod{p}?$$

Before formalizing this idea, let us build some intuition. Fix a prime p and a number a with $p \nmid a$. Consider the sequence

$$a, a^2, a^3, a^4, \dots \pmod{p}.$$

Since there are only finitely many nonzero residues modulo p , this sequence must eventually repeat. Moreover, by Fermat's Little Theorem, we know that $a^{p-1} \equiv 1 \pmod{p}$, so the value 1 must appear at some point.

The order of a modulo p is precisely the *first time* this sequence returns to 1. After that, the powers of a repeat in a cycle, and this cycle has length equal to the smallest exponent.

We now introduce the formal definition.

Definition. The **order** of a nonzero integer a modulo a prime p is the smallest positive integer k such that

$$a^k \equiv 1 \pmod{p}.$$

It is denoted by $\text{ord}_p(a)$.

Here's the first theorem regarding $\text{ord}_p(a)$, which is one of the most important lemmas in this section:

Lemma. If

$$a^X \equiv 1 \pmod{p}$$

then $\text{ord}_p(a) \mid X$.

Proof. This is a very nice application of the *Euclidean Algorithm*. Let $\text{ord}_p(a) = n$.

We now define the following:

$$X = nt + s, \quad \forall 0 \leq s < n$$

for positive integers t, s . Then, observe that

$$1 \equiv a^X \equiv a^{nt+s} \equiv (a^n)^t \cdot a^s \equiv (1)^t a^s \equiv a^s \pmod{p}$$

As $s < n$, we must have that $s = 0$. Otherwise, it would contradict the minimality of n . $\square$

Furthermore, convince yourself that if $a^X \equiv a^Y \equiv 1 \pmod{p}$, then

$$a^{\gcd(X,Y)} \equiv 1 \pmod{p}.$$

Comment. In fact, here's the lemma that we will use a lot of the time (try to see why this is true):

$$\text{ord}_p(a) \mid p - 1$$

In general, as well, whenever you have minimality of something, it is not too uncommon to make use of the *Euclidean Algorithm*.

Lemma. Let p be a prime. If for some a and b , $\text{ord}_p(a) = x$ and $\text{ord}_p(b) = y$ where x and y are coprime, then $\text{ord}_p(ab) = xy$.

Proof. Let $\text{ord}_p(ab) = t$. Then, by definition, t is minimal such that

$$(ab)^t \equiv 1 \pmod{p} \implies a^t \equiv b^{-t} \pmod{p}.$$

Raising both sides to the power of x , we get:

$$b^{-tx} \equiv a^{tx} \equiv (a^x)^t \equiv 1 \pmod{p} \implies b^{tx} \equiv 1 \pmod{p}$$

This gives us that $\text{ord}_p(b) = y \mid tx$. Making use of the coprime condition, we get that $y \mid t$. Similarly, we get that $x \mid t$ (verify this), so $xy \mid t$. Now, observe that

$$(ab)^{xy} \equiv (a^x)^y (b^y)^x \equiv 1 \pmod{p}$$

so xy works. Hence, we get that $\text{ord}_p(ab) = xy$. $\square$

Here's a classic problem trivialised by the above concept.

Exercise 3.1: Find all $n \in \mathbb{N}$ such that

$$n \mid 2^n - 1$$

Proof. This problem showcases how to use *orders*. If you try out the first 10 values of n , you will note that the only solution is $n = 1$, so that is what we would like to prove.

Firstly, we consider some prime divisor p of n . We get

$$p \mid n \mid 2^n - 1 \implies 2^n \equiv 1 \pmod{p}$$

Write $n = p^a \cdot r$, where $p \nmid r$. This would give us

$$2^{r \cdot p^a} \equiv (2^r)^{p^a} \equiv (2^r)^{p^{a-1}} \equiv \dots \equiv 2^r \equiv 1 \pmod{p}$$

Combining with that fact that $2^{p-1} \equiv 2^r \equiv 1 \pmod{p}$, we get

$$2^{\gcd(r, p-1)} \equiv 1 \pmod{p}$$

The best case scenario would be to show that $\gcd(r, p-1) = 1$ (remember, we want to show that $n = 1$, that is, get a contradiction). We want to choose a p such that $p-1$ shares no prime factors with r . If we choose p as the smallest prime divisor of n , then this would give us the require contradiction (why?) $\square$

We can now answer the question we originally asked:

Lemma. For an odd prime p , the following two statements are equivalent:

- a) p is of the form $4k + 1$
- b) p divides $n^2 + 1$ for some $n \in \mathbb{Z}$

Proof. Can you see why the second part implies the first? Observe that

$$n^2 + 1 \equiv 0 \pmod{p} \iff n^2 \equiv -1 \pmod{p} \implies n^4 \equiv 1 \pmod{p}$$

Now, note that $\text{ord}_p(n) \mid 4$, so

$$\text{ord}_p(n) = 1, 2 \text{ or } 4$$

If $\text{ord}_p(n) = 1$, then it will imply that $n \equiv 1 \pmod{p} \implies n^2 + 1 \equiv 2 \pmod{p}$, but this is only possible when $p = 2$; which doesn't work, as we are considering odd primes.

If $\text{ord}_p(n) = 2$, then observe that

$$n^2 \equiv 1 \not\equiv -1 \pmod{p}$$

which is again contradictory. In the end, we can conclude that $\text{ord}_p(n) = 4$. Now, observe that

$$4 = \text{ord}_p(n) \mid p - 1 \implies p \text{ is of the form } 4k + 1$$

For the second part, for a $4k + 1$ prime, can you construct some n such that $p \mid n^2 + 1$? (hint: consider $(p - 1)!$ and show that it is congruent to $\left(\frac{p-1}{2}!\right)^2$, which is a square modulo p) $\square$

We can push this even further, and prove Fermat's *Christmas Theorem*:

Theorem — Fermat's Christmas Theorem: Precisely the primes of the form $4k + 1$ can be written as $a^2 + b^2$, for some $a, b \in \mathbb{Z}^+$.

Proof. It is clear that p cannot be of the form $4k + 3$ due to a contradiction modulo 4. Using the previous lemma, consider n such that $p \mid n^2 + 1$. Now, use Thue's Lemma to find

$$1 \leq a, b < \sqrt{p} \text{ such that } a - bn \equiv 0 \pmod{p}$$

Now, note that

$$p \mid a^2 + b^2 < \sqrt{p}^2 + \sqrt{p}^2 = 2p$$

so $p = a^2 + b^2$ $\square$

We can do even more:

Lemma. A positive integer N can be written as a sum of two squares iff the power of all primes of the form $4k + 3$ dividing N is even.

Proof. For this, note that

$$\begin{aligned} (a^2 + b^2)(c^2 + d^2) &= a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 \\ &= (a^2c^2 + b^2d^2 + 2acbd) + (a^2d^2 + b^2c^2 - 2adbc) \\ &= (ac + bd)^2 + (ad - bc)^2 \end{aligned}$$

This is called *Brahmagupta's Identity*, and it tells us that if integers A, B are sums of two perfect squares, so is their product AB . (It is left to you as an exercise to convince yourself that this is enough for the proof). $\square$

Exercise 3.2: Show that exactly primes p of the form $3k + 1$ can be written as $a^2 + ab + b^2$ for some positive integers a, b .

Proof. You are encouraged to try this on your own. If you feel unsure about any step, you can try to replicate the proof for *Fermat's Christmas Theorem*.

Again, say for some integers a, b

$$p = a^2 + ab + b^2 \implies p \mid (a - b)(a^2 + ab + b^2) = a^3 - b^3$$

Note that a, b are smaller than p and hence are coprime with it. Now, we have

$$a^3 \equiv b^3 \pmod{p} \implies \left(\frac{a}{b}\right)^3 \equiv 1 \pmod{p}$$

In particular $\text{ord}_p\left(\frac{a}{b}\right) = 1, 3$. We will like to prove 1 doesn't work (why?). Note that if the order is 1, then that would give us that

$$\left(\frac{a}{b}\right) \equiv 1 \pmod{p} \iff p \mid a - b$$

which gives us a size contradiction (try to complete the contradiction for yourself). Hence, the order *has* to be 3, so $3 \mid p - 1$.

For the second part, consider t such that

$$t^3 \equiv 1 \pmod{p} \text{ and } t \not\equiv 1 \pmod{p}$$

Why will such a t exist? We will study that in the next chapter. For now, you may take it as granted.

Now, take $1 \leq a, b < \sqrt{p}$ using Thue's Lemma, such that

$$a - tb \equiv 0 \pmod{p} \iff p \mid a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Note that $p \nmid a - b$ so $p \mid a^2 + ab + b^2 < 3p$. So $a^2 + ab + b^2 = p$ or $2p$. Note that if both a, b are odd or if one is odd and the other is even, then $a^2 + ab + b^2$ is also odd, hence it can't be equal to $2p$. And if both a, b are even, then $4 \mid a^2 + ab + b^2 = 2p \implies 2 \mid p$ is also a contradiction. Hence, we get that $a^2 + ab + b^2 = p$. $\square$

We have looked at orders modulo a prime. Now let us try to generalize this:

Definition. For coprime positive integers m, n we define $\text{ord}_n(m) = k$ to be the smallest positive integer such that $m^k \equiv 1 \pmod{n}$

The same question arises: does m always exist? The answer is yes, this time because of Euler's theorem:

$$m^{\varphi(n)} \equiv 1 \pmod{n}$$

A lot of the same properties hold true here as well. Take this as an opportunity to try and prove these theorems by trying to replicate what we have done above:

Theorem — For coprime positive integers m, n , $\text{ord}_n(m) \mid \varphi(n)$

Make use of the *Euclidean Algorithm*, like we did earlier,

Theorem — For coprime positive integers m, n , if we have

$$m^a \equiv m^b \equiv 1 \pmod{n}$$

then $\text{ord}_n(m) \mid \gcd(a, b)$

With this, let us move on to the next exercise:

Exercise 3.3 (Nordic MO 2025/2): Let p be a prime and suppose

$$2^{2p} \equiv 1 \pmod{2p+1}$$

Prove that $2p+1$ is prime.

Proof. FTSOC assume that there exists some prime divisor q of $2p+1$. Then, observe that

$$2^{2p} \equiv 4^p \equiv 1 \pmod{q}$$

so $\text{ord}_q(4) = 1$ or p . From here, there are two cases:

Case 1: $\text{ord}_q(4) = p$

→ In this case, note that

$$\text{ord}_q(4) = p \mid q-1 \implies q \geq p+1$$

But then,

$$2p+1 \geq 2q \geq 2(p+1) = 2p+2$$

which is contradictory.

Case 2: $\text{ord}_q(4) = 1$

→ Then, $4^1 - 1 \equiv 0 \pmod{q} \iff q = 3$. Hence, 3 is the only prime divisor of $2p+1$ (as it is odd), say $2p+1 = 3^a$

Now, from the problem statement, we get

$$4^p \equiv 1 \pmod{3^a} \implies \text{ord}_{3^a}(4) = p \mid \varphi(3^a)$$

Recall that φ is the *Euler Totient Function*. Now, note that

$$p \mid \varphi(3^a) = 2 \cdot 3^{a-1} = \frac{2}{3} \cdot (2p+1)$$

But observe that $p < \frac{2}{3} \cdot (2p+1) < 2p$, which is a contradiction. □

Exercise 3.4 (STEMS 2025 Cat A Objective): Find the smallest natural number k such that there exist (possibly negative) integers $a_1, a_2, \dots, a_k$ such that

$$a_1^{11} + a_2^{11} + \dots + a_k^{11} = 2057$$

Proof. The main idea here is to find a prime of the form $p = 11n + 1$. We do this because

$$\varphi(p) = p - 1 = 11n$$

Then, note that

$$(a_i^{11})^n \equiv 0, 1 \pmod{p}$$

so if we solve the equation

$$X^n - 1 \pmod{p}$$

we will get all possible values of $a_i^{11} \pmod{p}$.

Now, we are quite lucky, since taking $n = 2$ gives us $11 \times 2 + 1 = 23$, a prime. So,

$$(a_i^{11})^2 \equiv a_i^{22} \equiv 0, 1 \pmod{23} \implies a_i^{11} \equiv 0, \pm 1 \pmod{23}$$

Now, observe that for $k \leq 9$,

$$-k \leq \sum_{i=1}^k a_i^{11} \leq k \pmod{23},$$

so it can never be equal to $2057 \equiv 10 \pmod{23}$. Hence $k \geq 10$. Now, for 10, note that

$$2^{11} + \underbrace{1^{11} + 1^{11} + \dots + 1^{11}}_{9 \text{ times}} = 2057.$$

□

3.1.2 Types of Primes

Primes have been objects of mathematical wonder which go back as far as the origin of mathematics itself. It is only natural to ponder upon the existence of primes of a certain form. For example, the infinitude of primes in an arithmetic progression with the first two terms coprime has been already established by Dirichlet. However, the infinitude of primes of a particular quadratic form is not yet known, not even primes of the form $n^2 + 1$. Having studied properties of orders, we shall now restrict our attention to the primes which are one away from powers of two.

Fermat Primes: Primes of the form $2^N + 1$ are called *Fermat Primes*. Despite having covered nothing but the definition, here is something you can already prove:

Lemma. If $2^N + 1$ is a prime, then N itself must be a power of 2.

Proof. Suppose not. Then there exist some odd prime p dividing N . Say, $N = pm$ for some positive m . Then note that

$$2^m + 1 \mid (2^m)^p + 1 = 2^{pm} + 1 = 2^N + 1$$

Indeed, this is the definition you will see in most places: primes of the form $2^{2^n} + 1$. This function grows extremely fast, taking n as merely 10 will give us a number with 309 digits. The largest known Fermat prime is $2^{2^4} + 1 = 65537$, which is just 5 digits. In fact, it is conjectured that only a finite number of Fermat primes exist. □

In general, number of the form $2^{2^n} + 1, n \in \mathbb{N}$ is called a *Fermat number*. A very important lemma about prime divisors of Fermat number goes as follows:

Lemma. Any prime divisor p of $2^{2^n} + 1$ is of the form $2^{n+1}k + 1$.

Proof. Suppose $p \mid 2^{2^n} + 1$. Then $2^{2^n} \equiv -1 \pmod{p} \implies 2^{2^{n+1}} \equiv 1 \pmod{p}$. So $\text{ord}_p(2) \mid 2^{n+1}$. But note that the order cannot divide 2^n (why?), so it must be exactly 2^{n+1} . Hence, seeing that $2^{n+1} = \text{ord}_p(2) \mid p - 1$, we are done. $\square$

Besides their neat form, Fermat primes also provide links between Number theory and Geometry because of the following result due to Gauss and Wantzel:

Theorem — Gauss-Wantzel Theorem: A regular polygon with n sides can be constructed using only a ruler and a compass iff every odd prime divisor of n is a Fermat prime.

The proof, of course, is quite non-elementary and requires a ton of higher mathematics, hence we shall omit it.

Using this theorem, we get that a regular polygon with

$$4294967295 = 3 \times 5 \times 17 \times 257 \times 65537$$

sides can be constructed. This is also the largest odd number which we know to be constructible.

Mersenne Primes: We looked at primes one above a power of 2, let us now look at those which are one below a power of 2. They are known as *Mersenne Primes*, names after French polymath *Marin Mersenne*, whose heavy work popularised these primes.

As earlier, there is something that we can already prove:

Lemma. If $2^n - 1$ is prime, then n is prime as well.

Proof. FTSOC assume the contrary, and consider a divisor $1 < d < n$ of n . Suppose $n = dk$. Simply observe that

$$2^n - 1 = 2^{dk} - 1 = (2^d - 1)(2^{d(k-1)} + 2^{d(k-2)} + \dots + 1)$$

which contradicts its primality, since $2^d - 1$ divides a prime. $\square$

Notice how this is not an *if and only if* condition. For a quite a while, it was believed that $2^p - 1$ is prime for all primes p , and for good reason too: $p = 2, 3, 5, 7, 11, 13, 17, 19, 23$ all satisfy the primality. However, it was Mersenne himself who disproved the conjecture by showing that $2^{29} - 1$ is composite

$$2^{29} - 1 = 536870911 = 233 \times 1103 \times 2089$$

The infinitude of Mersenne primes is not yet known, however, whenever a new Mersenne prime is discovered, it is quite often the largest prime known at the time. Currently, it is

$$2^{136279841} - 1 \approx 10^{41024320}$$

Thats right, it has over **forty million** digits, which the human mind can hardly comprehend.

You might be wondering how it is even possible to confirm whether a number with millions and millions of digits is prime. A crude way to check if a number is prime is to just check if it is divisible by any of the numbers less than it. That would take more time than the universe has existed, so this is not at all viable. A slightly smarter idea would be to check its divisibility by only primes upto $\sqrt{n}$; however this too will take far too long, not even including the fact that we do not have a complete list of all primes till such a large number. Instead, a primality test was used, known as the **Lucas-Lehmer Primality Test for Mersenne Primes**, abbreviated as ‘LLT’. The test states the following:

Theorem — Lucas-Lehmer Primality Test: Consider the sequence of positive integers defined as follows:

$$a_1 = 4, \quad a_n = a_{n-1}^2 - 2$$

The test states that for any prime p , $2^p - 1 \mid a_{p-1}$ iff $2^p - 1$ is a Mersenne prime.

Though the theorem may seem extremely random, the proof is actually within our reach, provided we make use of a little bit of Algebraic Number Theory. We shall cover it at the end of this section.

Contrary to the Fermat Primes, it is strongly believed by many mathematicians that infinitely many Mersenne primes exist. Of course, this is still unsolved, but there does exist a conjecture to heuristically estimate the number of Mersenne primes less than or equal to a given quantity: The **Lenstra-Pomerance-Wagstaff** states that asymptotically,

$$\pi_M(X) = \frac{e^\gamma \log \log X}{\log 2}$$

where $\pi_M(X)$ = number of Mersenne primes less than or equal to X . e is just Euler’s number, but γ is the **Euler-Mashceroni Constant** defined as follows:

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \frac{1}{i} - \int_1^n \frac{1}{x} dx \right) = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n \frac{1}{i} - \log n \right) \approx 0.57721 \dots$$

Mersenne primes are deeply inter-related with what are known as *Perfect Numbers*. A positive integer N is said to be a Perfect Number if it is exactly equal to the sum of

its divisors excluding the number itself. The first few perfect numbers are:

$$\begin{aligned} 6 &= 1 + 2 + 3 \\ 28 &= 1 + 2 + 4 + 7 + 14 \\ 496 &= 1 + 2 + 4 + 8 + 16 + 31 + 62 + 124 + 248 \end{aligned}$$

and so on. Euclid was the first one to make any real progress on this.

Lemma. (*Due to Euclid*) Any number of the form

$$2^{p-1}(2^p - 1)$$

is perfect, given that both p and $2^p - 1$ are primes.

Proof. The proof is quite simple, observe that the divisors of this number are precisely:

$$\{1, 2, \dots, 2^{p-1}\} \quad \text{and} \quad \{(2^p - 1)1, (2^p - 1)2, \dots, (2^p - 1)2^{p-1}\}.$$

Consider the sum of the all divisors excluding the last:

$$(1 + 2 + \dots + 2^{p-1}) + (2^p - 1)(1 + 2 + \dots + 2^{p-2}) = 2^p - 1 + (2^p - 1)(2^{p-1} - 1) = (2^p - 1)(2^{p-1})$$

□

What Euclid couldn't do was prove that every even perfect number was of the above form. In fact, it remained unproven for about 2000 years after Euclid, until we got Euler in the picture. Euler invented a new weapon, the sigma function (σ) for the sum of divisors, which we have studied in chapter 2.

Lemma. (*Due to Euler*) All even perfect numbers are of Euclid's form.

Proof. Suppose N is an even perfect number. Then N is equal to the sum of its divisor excluding itself, or in other words,

$$\sigma(N) = 2N$$

The idea is that we would expect $\sigma(n)$ to overshoot in many cases. Suppose $N = 2^a b$, where b is odd. Recall that σ is multiplicative, that is, $\sigma(ab) = \sigma(a)\sigma(b)$ for coprime a, b . Using this, we get

$$2N = \sigma(N) = \sigma(2^a b) = \sigma(2^a)\sigma(b)$$

We know that $\sigma(2^a) = 1 + 2 + \dots + 2^a = 2^{a+1} - 1$. So we have

$$(2^{a+1} - 1)\sigma(b) = 2^{a+1}b \implies 2^{a+1} - 1 \mid b$$

This implies

$$\sigma(b) = 2^{a+1} \cdot \frac{b}{2^{a+1} - 1}$$

Observe that because $2^{a+1} - 1$ divides b , we must have that $\frac{b}{2^{a+1}-1}$ is a divisor b . Further, this divisor is not b as the denominator is not 1. Therefore, we get that

$$2^{a+1} \frac{b}{2^{a+1} - 1} = \sigma(b) \geq b + \frac{b}{2^{a+1} - 1} = \frac{2^{a+1}b}{2^{a+1} - 1}$$

Aha! Equality holds, which is only possible when there is no other divisor. So $b = 2^{a+1} - 1$, and further, b is prime, which is what we required. $\square$

This tells us that there are infinitely many even perfect numbers iff there are infinitely many Mersenne primes. What about odd perfect numbers? The investigation of odd perfect numbers is much harder, and not much is known about them. Rene Descartes hypothesized that odd perfect numbers have to be of a specific form. However, he was unable to prove his hypothesis. It was later proven by, of course, none other than Euler:

Lemma. (Due to Euler, again) Suppose N is an odd perfect number. Then

$$N = pm^2$$

where p is an odd prime of the form $4k + 1$

Proof. Euler's sigma function was used to attack this as well. We know that $\sigma(N) = 2N$. Write $N = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$. Recall that

$$\sigma(N) = \sigma(p_1^{\alpha_1}) \cdots \sigma(p_k^{\alpha_k})$$

Note that if α_i is odd then $\sigma(p_i^{\alpha_i})$ is even. Observe that the maximum power of 2 dividing $2N$ is 2^1 , so that is also the maximum power of 2 dividing

$$\sigma(N) = \sigma(p_1^{\alpha_1}) \cdots \sigma(p_k^{\alpha_k})$$

Therefore, exactly one of the $\sigma(p_i^{\alpha_i})$'s is even and the rest are odd, that is, only 1 α_i , say α_t is odd and the rest are even, which proves the claim. Further, see that if p_t is of the form $-1 \pmod{4}$, then $\sigma(p_t^{\alpha_t})$ is divisible by 4 (why?), so this isn't possible, as we just showed that $\sigma(N)$ cannot be divisible by 4. So p_t is of the form $4k + 1$, and we can write, for $m = p_1^{a_1/2} \cdots p_t^{a_t-1/2} \cdots p_k^{a_k/2}$, $N = p_t m^2$. $\square$

This is where Euler left the scene, and entered English Mathematician James Joseph Sylvester. He proved the following in less than a year:

Theorem — (Due to J.J. Sylvester) Suppose N is an odd perfect number. Then,

1. N cannot be divisible by 105
2. N must have atleast 4 distinct prime divisors

3. If N isn't divisible by 3, then it has atleast 8 distinct prime factors

We will be proving only the first one. To generalize the notion of perfect numbers, Sylvester invented what is known as the *Abundancy Index* $I(n)$, defined as

$$I(n) = \frac{\sigma(n)}{n}$$

For our purposes, N is perfect when $I(n) = 2$. Sylvester proved the following:

Lemma. For every coprime $k, n \in \mathbb{Z}$, $I(kn) > I(n)$

Proof. Observe that, if $d_1, d_2, \dots, d_l$ are divisors of n , then $kd_1, \dots, kd_l$ are divisors of kn .

$$\begin{aligned} I(kn) &= \frac{\sigma(kn)}{kn} \\ &> \frac{kd_1 + \dots + kd_l + \sigma(n)}{kn} \\ &= \frac{(k+1)\sigma(n)}{kn} \\ &> \frac{\sigma(n)}{n} = I(n) \end{aligned}$$

□

Exercise 3.5: There are no perfect numbers divisible by 105.

Proof. From this suppose $N = 3^a \cdot 5^b \cdot 7^c k$, where k is coprime and $a, b, c \geq 1$. See that 3 and 7 are primes of the form $4k - 1$, so their powers are atleast 2. Hence,

$$\begin{aligned} I(N) &= I(3^a 5^b 7^c k) \\ &\geq I(3^a 5^b 7^c) \\ &= I(3^a) I(5^b) I(7^c) \\ &= \left(\frac{3^{a+1} - 1}{3 - 1} \right) \left(\frac{5^{b+1} - 1}{5 - 1} \right) \left(\frac{7^{c+1} - 1}{7 - 1} \right) \\ &= \left(\frac{3^{a+1} - 1}{2 \cdot 3^a} \right) \left(\frac{5^{b+1} - 1}{4 \cdot 5^b} \right) \left(\frac{7^{c+1} - 1}{6 \cdot 7^c} \right) \\ &\geq \left(\frac{3^3 - 1}{2 \cdot 3^2} \right) \left(\frac{5^2 - 1}{4 \cdot 5^1} \right) \left(\frac{7^3 - 1}{6 \cdot 7^2} \right) > 2 \end{aligned}$$

so N cannot be a perfect number. □

Now, let us look at some **Solved Examples**:

Solved Example

3.1 (Well-known): Fix any positive integer n . Show that there are infinitely many primes of the form $k \cdot 2^n + 1$.

Solution: Let us just forget for a moment that Dirichlet's Theorem exists. We want to find some N such that $\varphi(N)$ is a power of 2. That's right, if we just let N be a power of 2.

Consider a prime divisor of $X^{2^{N-1}} + 1$.

$$X^{2^{N-1}} \equiv -1 \pmod{p} \implies \text{ord}_p(X) = 2^N \mid p - 1$$

So we are done! Or are we? What if there are only finitely many primes dividing $X^{2^{N-1}} + 1$? There are many theorems which can kill this, like *Schur's Lemma*, but let us try to prove this.

FTSOC suppose that there are only finitely many prime divisors of $X^{2^{N-1}} + 1$. Call these $p_1, p_2, \dots, p_k$. Let $X = p_1 p_2 \cdots p_k$. (see why this finishes).

Solved Example

3.2 (Putnam 1994/B6): For $a \in \mathbb{Z}$ define

$$n_a = 101a - 100 \cdot 2^a$$

show that, for $0 \leq a, b, c, d \leq 99$ we have

$$n_a + n_b \equiv n_c + n_d \pmod{10100} \implies \{a, b\} = \{c, d\}$$

Solution: See that the required condition boils down to

$$n_a + n_b \equiv n_c + n_d \pmod{100} \text{ and } n_a + n_b \equiv n_c + n_d \pmod{101}$$

due to CRT. By doing modulo 100 we get,

$$101a - 100 \cdot 2^a + 101b - 100 \cdot 2^b \equiv 101c - 100 \cdot 2^c + 101d - 100 \cdot 2^d \pmod{100}$$

$$\implies a + b \equiv c + d \pmod{100}$$

And by using mod 101 we get,

$$2^a + 2^b \equiv 2^c + 2^d \pmod{101}$$

Now using Fermat's Little Theorem we get,

$$2^{a+b} \equiv 2^{c+d} \pmod{101} \implies 2^a \cdot 2^b \equiv 2^c \cdot 2^d \pmod{101}$$

Note from both the conditions we get

$$\{2^a, 2^b\} = \{2^c, 2^d\}$$

as we have $a, b, c, d \leq 99$ we just need to show that 2 is a primitive root modulo 101. So, we just need to show $2^{20}, 2^{50} \not\equiv 1 \pmod{101}$.

$$2^{20} \equiv 1024^2 \equiv 14^2 \equiv 196 \equiv 95 \equiv -6 \pmod{101},$$

$$2^{50} \equiv 1024^5 \equiv 14^5 \equiv 14 \cdot 196^2 \equiv 14 \cdot (-6)^2 \equiv 14 \cdot 36 \equiv 504 \equiv -1 \pmod{101}.$$

Now, we have proved 2 is a primitive root, hence this finishes.

Solved Example

3.3 (USA TST 2008/4): Prove that for no integer n is $n^7 + 7$ a perfect square.

Solution: First, we check the equation modulo 4 to get that n is odd. Also, always remember the rule of thumb: *Always* check the first few values of n . Even though we have to prove that there exists no such n , it will give us a good idea on what fails. If you did check a few values, you will notice that $n = -2$ is *almost* a solution:

$$(-2)^7 + 7 = -(11^2)$$

but we want $+11^2$. Nonetheless, this is still useful information: we can write 7 in this way and get

$$n^7 + 7 = m^2 \iff n^7 + 2^7 = (n + 2)(n^6 - 2n^5 + \dots + 2^6) = m^2 + 11^2$$

Aha! RHS is the sum of two squares, so we may invoke the generalization of Fermat's *Christmas Theorem*. But we need to be careful. We can't say right away that any prime divisor p of $m^2 + 11^2$ is of the form $4k + 1$; what if $11 \mid m$ or m is odd? So we split into two cases:

Case 1: $11 \nmid m$, and m is odd.

→ This is simple enough, and we can directly invoke the theorem to say that if odd prime

$$p \mid m^2 + 11^2 \implies p \equiv 1 \pmod{4}$$

Now, note that

$$n + 2 \mid m^2 + 11^2 \implies n + 2 \equiv 1 \pmod{4} \iff n \equiv -1 \pmod{4}$$

Not then, we get

$$n^7 + 7 \equiv (-1)^7 - 1 \equiv 2 \pmod{4}$$

and $m^2 \equiv 2 \pmod{4}$ is impossible (why?)

Case 2: $11 \mid m$, and m is odd.

→ Then let $m = 11r$. Then, the RHS would become

$$11^2 p^2 + 11^2 = 11^2(p^2 + 1)$$

From the previous case, $n + 2$ cannot be $1 \pmod{4}$. Observe that 11 is the only $4k+3$ prime dividing the RHS, so 11 but not 121 divides $n+2$. Then $n \equiv -2 \pmod{11}$.

However, we also have that:

$$\begin{aligned} \frac{n^7 + 2^7}{n + 2} &= n^6 - 2n^5 + 4n^4 - 8n^3 + 16n^2 - 32n + 64 \\ &\equiv \underbrace{64 + 64 + \cdots + 64}_{7 \text{ times}} \\ &\equiv 7 \times 64 \\ &\equiv 8 \pmod{11} \end{aligned}$$

Hence, 11 would divide the LHS only once, which is a contradiction as $11^2 \mid 11^2(p^2 + 1)$.

The case where m is even is very similar to the above case (in fact a bit simpler), so we encourage you to finish this case (look at when 11 divides m and when it doesn't).

Solved Example

3.4 (IMO 1990/3): Determine all integers $n > 1$ such that

$$\frac{2^n + 1}{n^2}$$

is an integer.

Solution: The most common first step is to consider primes, so let p be a prime divisor of n . We can work with any prime divisor for now, and later add constraints, so as to extract some useful information: Let $n = p^a k$, where $p \nmid k$. So we get

$$2^n \equiv 2^{p^a k} \equiv -1 \pmod{p} \implies 2^{2p^a k} \equiv 2^{2k} \equiv 1 \pmod{p}$$

In particular, we have that

$$\text{ord}_p(2) \mid 2k \text{ and } p-1 \implies \text{ord}_p(2) \mid \gcd(2k, p-1) = 2 \gcd\left(k, \frac{p-1}{2}\right)$$

Now, we will do two things: First, we put p as the smallest prime divisor of n . Then, note that all the divisors of k is greater than p , whereas those of $\frac{p-1}{2}$ are less than p .

Hence the two are coprime, giving us that:

$$2^2 \equiv 1 \pmod{p} \implies p \mid 3 \implies p = 3 \text{ (also, } 2^1 \not\equiv 1 \pmod{3})$$

Now, we put p as the second smallest prime divisor of n . Then note that the prime divisors of k is either 3 or greater than p . On the other hand, the prime divisors of $\frac{p-1}{2}$ is less than p . Therefore, their greatest common divisor is a power of 3, say 3^b .

Now, we have

$$2^{2 \cdot 3^b} \equiv 1 \pmod{p} \text{ but } 2^{3^b} \equiv -1 \pmod{p}$$

As a result, $\text{ord}_p(2) \mid 2 \cdot 3^b$ but not 3^b , so the only possible case is that $\text{ord}_p(2) = 2 \cdot 3^b$; but b can only be 0 or 1. Try to find yourself why $b \not\geq 2$ after reading the full solution. However, if the order is 6, then

$$p \mid 2^6 - 1 = 7 \cdot 3^2$$

However, see that p cannot be 7 (try to see why by using the fact that the power of n is 2) which again gives us that

$$2^2 \equiv 1 \pmod{p} \implies p \mid 3$$

Here's the thing though: the second smallest prime divisor is clearly greater than the smallest prime divisor. So we can use this to conclude that 3 is the only prime divisor of n , or, in other words, n is a power of 3, say 3^c . This gives us

$$3^{2^c} \mid 2^{3^c} + 1$$

Now, ideally, you would finish using *LTE*. However, we will be covering that not until the next chapter (Further, the participants of the IMO likely did not know about *LTE* at the time of the contest. Hence, we should learn to deal with this in another way too.)

We shall finish this using an induction. checking a few values of c tells us that $c = 0, 1$ yield the only solutions, so we want to show that for $c \geq 2$ there exists no solution.

For the base case, note that

$$3^{2 \cdot 2} = 81 \nmid 2^{3^2} + 1 = 2^9 + 1 = 513$$

Now, for the induction hypothesis, assume that

$$3^{2^d} \nmid 2^{3^d} + 1$$

for some $d \geq 2$. Observe that

$$\begin{aligned} 2^{3^{d+1}} + 1 &= 2^{3^d \cdot 3} + 1 \\ &= \left(2^{3^d}\right)^3 + 1 \\ &= (2^{3^d} + 1)(2^{2 \cdot 3^d} - 2^{3^d} + 1) \end{aligned}$$

Observe that out of the two terms, the second term is divisible by 3 but not 9. This is because

$$2^{2 \cdot 3^d} - 2^{3^d} + 1 \equiv 1 - (-1) + 1 \pmod{9}$$

Hence, if

$$3^{2(d+1)} \mid 2^{3^{d+1}} + 1 = (2^{3^d} + 1)(2^{2 \cdot 3^d} - 2^{3^d} + 1)$$

were to hold then, it would imply

$$3^{2d+1} \mid 2^{3^d} + 1$$

which is contradictory.

Solved Example

3.5 (All Russia National Math Olympiad Grade 9 2025/3): Find all natural numbers n for which there exists an even natural number a such that the number

$$(a-1)(a^2-1) \cdots (a^n-1)$$

is a perfect square.

Solution: As always, if you try a few base cases, you will realize that $n = 1$ and $n = 2$ yield the only solutions. So this is what we should try to prove. Now, consider a prime divisor p of this bulky expression. Then, p divides at least one out of

$$a-1, a^2-1, \dots, a^n-1$$

Then, observe that $\text{ord}_p(a) \leq n$, and that the highest power to which p divides the expression must be even. It would be nice to find some $1 \leq r \leq n$ two things:

- a) We have a prime divisor p of *only* $a^r - 1$ and no other term. This is essentially the same as saying that $\text{ord}_p(a) = r$, and $\frac{n}{2} < r$
- b) p but not p^2 divides $a^r - 1$.

The break through is to consider the highest power of 2 less than n . So consider k such that $2^k \leq n < 2^{k+1}$. Now, consider

$$a^{2^k} - 1 = (a^{2^{k-1}} - 1)(a^{2^{k-1}} + 1)$$

Take an odd prime divisor p of $a^{2^{k-1}} + 1$ (We can do this as a is even). Notice it does not divide $a^{2^{k-1}} - 1$. Further, note that $a^{2^{k-1}}$ is a perfect square, so $a^{2^{k-1}} + 1$ isn't. Hence, there exists some odd prime divisor p whose maximum power dividing $a^{2^k} - 1$ is odd. Finally, see that

$$a^{2^{k-1}} \equiv -1 \pmod{p} \implies \text{ord}_p(a) \text{ divides } 2^k \text{ but not } 2^{k-1}$$

so the order can be only 2^k . Voilà! Both the conditions are satisfied, but only when $k \geq 2$, that is $n \geq 4$. We know that $n = 1, 2$ does yield solutions, so we want to disprove the case $n = 3$. Particularly, we want to show that

$$(a-1)(a^2-1)(a^3-1) = (a-1)^3(a+1)(a^2+a+1) = b^2$$

has no solutions. Note that this is the same as disproving

$$(a-1)(a+1)(a^2+a+1) = c^2$$

Consider a prime divisor q of $a+1$. Then it cannot divide $a-1$ or a^2+a+1 (as $q \neq 2$). Because $q \mid c^2$, the power of q dividing $a+1$ must be even, so $a+1$ is a perfect square, say d^2 . Now observe that we can say $\gcd(a-1, a^2+a+1) = 1$ if $3 \nmid a-1$ but that is obviously true as then

$$a+1 \equiv 2 \pmod{3}$$

but then it can't be perfect square. So, we will get that $a-1 = e^2$; but this is just contradiction as both $a+1, a-1$ are perfect squares. So, we are basically done!

Practice Problems:

3.1 (MMOS Entrance 2025/1): Find all pairs of positive integers m, n such that $3^m + 1$ and $3^n + 1$ are both divisible by mn .

3.2 (AoPS): Let p be a prime such that $2p+1 \mid 2^p - 1$. Then show that $2p+1$ is also a prime.

3.3 (Romania TST 1996/11): Find all primes p, q such that $\alpha^{3pq} - \alpha \equiv 0 \pmod{3pq}$ for all integers α .

3.4 (JBMOSL 2024/N2): Find all the pairs of $(p; q)$ distinct prime numbers such that such that

$$q^p \mid p + p^q + p^{q^p}$$

3.5 (Krishna Pothapragada): Let n be an even integer, and let $d(n)$ be the number of divisors of n . (where $d(n)$ is the number of divisors of n) Prove that

$$d(n) \nmid n^2 + 1$$

3.6 (Catherine Xu): Find all pairs (m, n) of positive integers such that $m^2 + 3 = n^3$.

3.7 (USA TST 2003/3): Find all ordered triples of primes (p, q, r) such that

$$p \mid q^r + 1, \quad q \mid r^p + 1, \quad r \mid p^q + 1.$$

3.8 (ISL 2017/N5): Find all pairs (p, q) of prime numbers which $p > q$ and

$$\frac{(p+q)^{p+q}(p-q)^{p-q} - 1}{(p+q)^{p-q}(p-q)^{p+q} - 1}$$

is an integer.

3.9 (Swedish Correspondence Course 21/22): The numbers m and n are positive integers, and the number p such that $p > n$ is a prime. We are given that

$$pm + n \mid p^p + 1$$

Prove that $n \mid m$.

Bonus Lemma: We shall see a proof of the **Lucas-Lehmer Primality Test**. This is best suited for readers who have a basic understanding of rings. If you do not know what they are, then you can either read up a bit on them or read this section anyways; we do include brief explanations. Keep in mind that this is optional (hence the name *bonus* lemma), and is not tested in contests.

Theorem — Lucas-Lehmer Primality Test: Consider the sequence of positive integers defined as follows:

$$a_1 = 4, \quad a_n = a_{n-1}^2 - 2$$

The test states that for any prime p , $2^p - 1 \mid a_{p-1}$ iff $2^p - 1$ is a Mersenne prime

We will only prove the ‘if’ part. Firstly, we define 2 numbers X and Y as follows: Let

$$X = 2 + \sqrt{3} \quad \text{and} \quad Y = 2 - \sqrt{3}$$

Observe that $XY = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$. Defining these might seem random at first glance, but they are quite well motivated due to the following lemma:

Lemma.

$$a_n = X^{2^{n-1}} + Y^{2^{n-1}}$$

Proof. Approaching this with an induction is quite natural. For the base case, $a_1 = 4 = 2 + \sqrt{3} + 2 - \sqrt{3} = X^{2^0} + Y^{2^0}$. Suppose this is true for some k . Then, note that

$$\begin{aligned} a_{k+1} &= a_k^2 - 2 \\ &= \left(X^{2^{k-1}} + Y^{2^{k-1}} \right)^2 - 2 \\ &= X^{2^k} + Y^{2^k} + 2X^{2^{k-1}}Y^{2^{k-1}} - 2 \\ &= X^{2^k} + Y^{2^k} + 2 - 2 \\ &= X^{2^k} + Y^{2^k} \end{aligned}$$

□

This was precisely the idea of defining X and Y . Writing the sequence in this form allows us to make use orders very well, just recall **Solved Example 3.1**.

However, the issue is that X and Y are no longer integers, so how exactly do we define ‘ $31 + 5\sqrt{3} \pmod{6}$ ’ or ‘ $20 + 2\sqrt{3} \pmod{5}$ ’? The idea is to work in the ring

$$\mathbb{Z}[\sqrt{3}] = \{a + b\sqrt{3} : a, b \in \mathbb{Z}\}$$

Recall what modular arithmetic meant in the most basic sense. In short, considering $(\text{mod } n)$ was indirectly saying $n = 0$. Let us put this exact meaning: for an integer $m \geq 1$, define

$$a + b\sqrt{3} \pmod{m} = (a \pmod{m}) + (b \pmod{m})\sqrt{3}$$

Essentially, we have taken the rational and irrational parts separately and considered them modulo m . Here people come across a common misconception: doesn't $\sqrt{3} \pmod{m}$ refer to an element x such that $x^2 \equiv 3 \pmod{m}$?

Indeed, this would have been true if we were considering the elements of $\mathbb{Z}$ modulo m , and since $\sqrt{3}$ doesn't exist in $\mathbb{Z}$, we would have had to define $\sqrt{3}$ as the number which squared gives 3. We could not have taken $\sqrt{3} \pmod{m}$ for any random m without certain conditions, since $x^2 \equiv 3 \pmod{m}$ does not always have solutions.

So what are we doing here? We are considering the elements of $\mathbb{Z}[\sqrt{3}]$ modulo m . Now observe that $\sqrt{3} \in \mathbb{Z}[\sqrt{3}]$ ($\sqrt{3} = 0 + 1 \cdot \sqrt{3}$). So we do not have to redefine what $\sqrt{3}$ is. It is just like considering $5 \pmod{7}$.

So then what exactly is the ' $\sqrt{3}$ ' in $\mathbb{Z}[\sqrt{3}]$? It's *not* representing $1.732\dots$. It doesn't represent the (positive) real number which squared gives 3. Of course, there are ways to argue that if you take a certain type of mapping from $\mathbb{Z}[\sqrt{3}]$ to $\mathbb{Z}$, then it behaves like $1.732\dots$, but in $\mathbb{Z}[\sqrt{3}]$ it more represents the number theoretic properties of the number $\sqrt{3}$ than its numeric value. It is very much analogous to how we write complex numbers as $a + bi$ and don't question the value of i .

Okay, that was a long digression. Let us come back to the primality test. Now, take a mersenne number $M_p = 2^p - 1$ and suppose it divides a_{p-1} and is composite. Say

$$\begin{aligned} a_{p-1} = X^{2^{p-2}} + Y^{2^{p-2}} = M_p k &\iff X^{2^{p-2}} X^{2^{p-2}} = (M_p k - Y^{2^{p-2}}) X^{2^{p-2}} \\ X^{2^{p-1}} = k M_p X^{2^{p-2}} - (XY)^{2^{p-2}} &= k M_p X^{2^{p-2}} - 1 \end{aligned}$$

Define q as the smallest prime divisor of M_p . Clearly M_p is not even, so $q > 2$. We take this equation modulo q , so we get

$$X^{2^{p-1}} \equiv -1 \pmod{q}$$

This is where the notion of modular arithmetic in $\mathbb{Z}[\sqrt{3}]$ which we developed earlier comes in. This time we get kind of lucky, as the q that we have considered happens to be an integer. So we are reducing both the rational and irrational part of X modulo q .

This equation is also very helpful as we can easily deduce that

$$\text{ord}_q(X) \mid 2^p, \text{ord}_q(X) \nmid 2^{p-1} \implies \text{ord}_q(X) = 2^p$$

We do have to be a bit careful now: Note that there are now q^2 elements modulo q , as there are 2 degrees of freedom, the rational and the irrational parts. Therefore $\text{ord}_q(X) \mid q^2 - 1$. And now for the final reveal:

$$2^p \leq q^2 - 1 < q^2 \leq M_p = 2^p - 1$$

and we have a contradiction. (Again, you might wonder if orders or Fermat's Little Theorem works when we consider $\mathbb{Z}[\sqrt{3}]$ modulo m . The answer is yes, it does. You can take this as an opportunity to try and work it out on your own by replicating our proofs for regular modulo.)